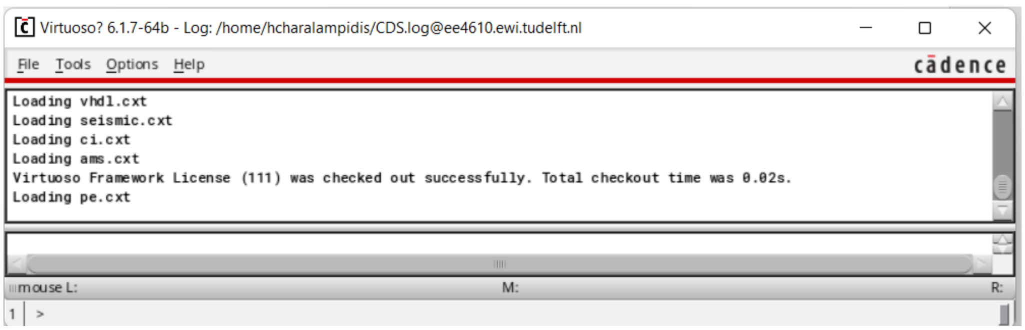
Cadence First Use Tutorial

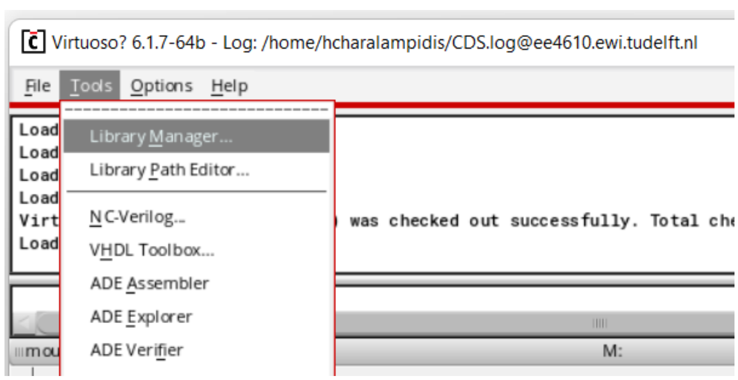
Rev. 20/02/23

1. In order to launch Cadence, check the guide named “EE4615_Instruction_Manual-1” on Brightspace. It is found under “Practical Information 2023”.

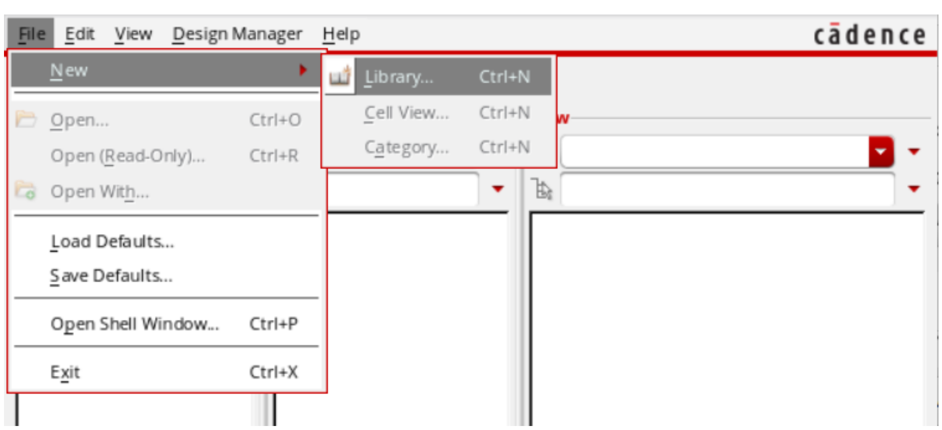
   After launching, the following window will pop up:



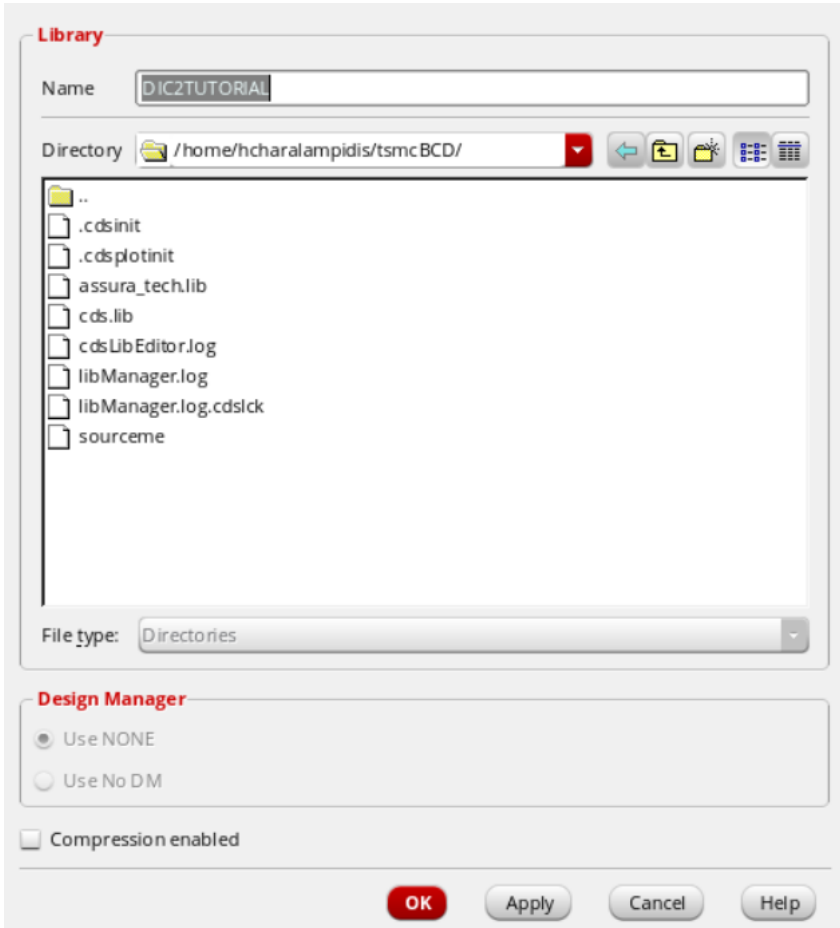
2. Before you can start working, you must first create a library. Go to Tools → Library Manager.



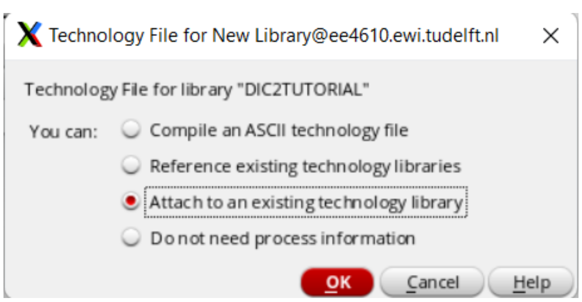
3. On the window that pops up, go to File → New → Library.

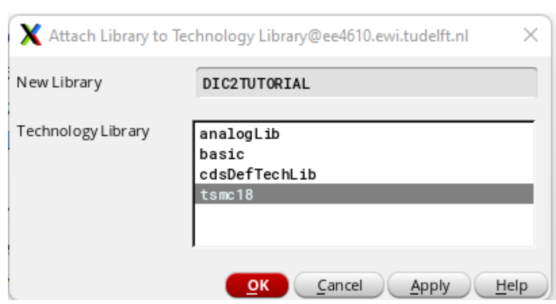


4. Pick a name for your library. Make sure you are in directory /tsmcBCD/. Press “OK”.



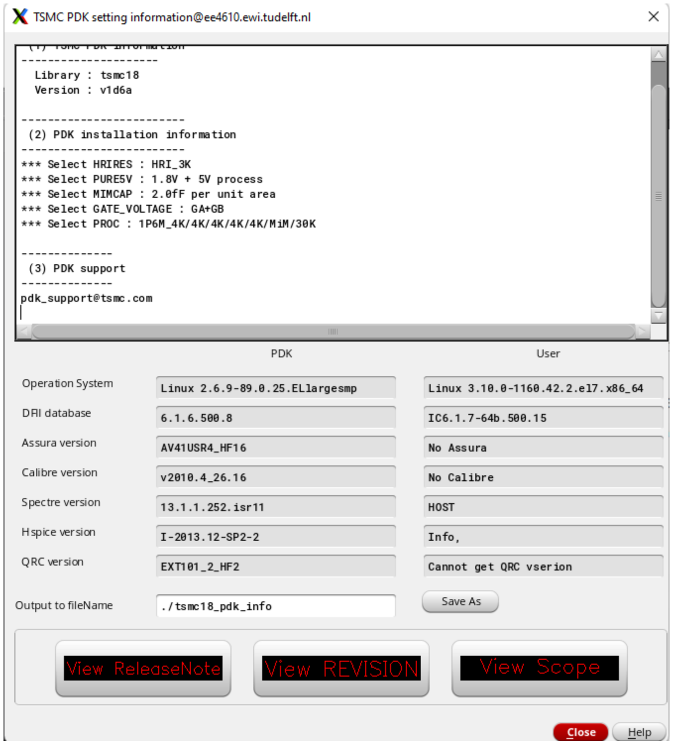
5. **Important:** The following window will pop up: “Technology File for library ...”. Pick the third option “Attach to an existing technology library”.

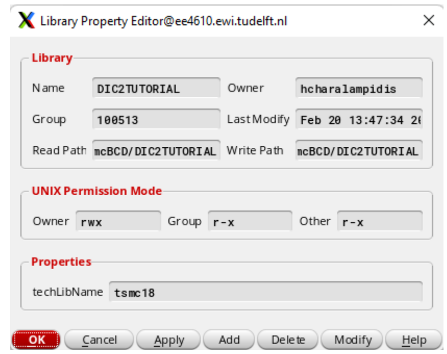


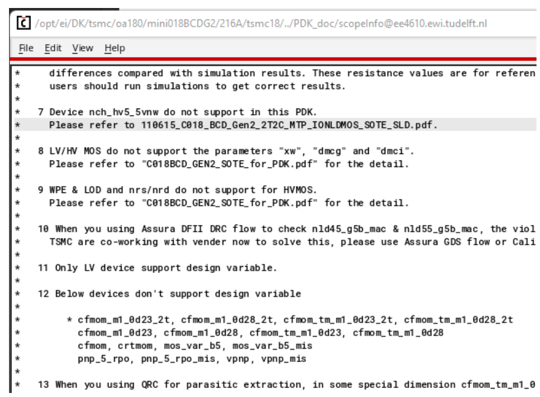


6. From the “Technology Library” options, pick “tsmc18”. This is the TSMC 180 nm technology that we are going to be using.

7. For the following pop-up windows, click “Close”, “OK”, or “X”.

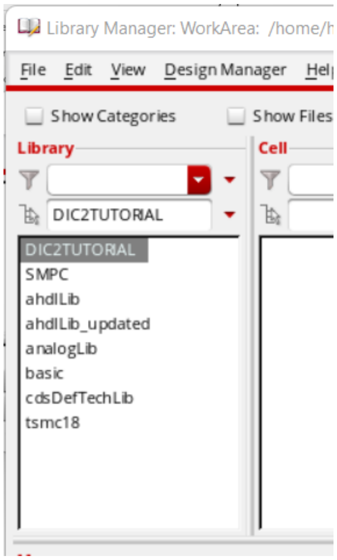


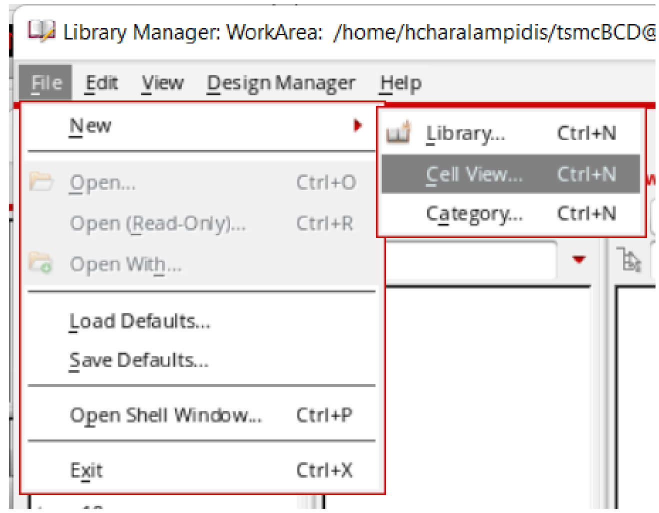




8. Now you must create your main cell. Your “Library Manager” window should still be open. If not, check step 2 on how to open it.

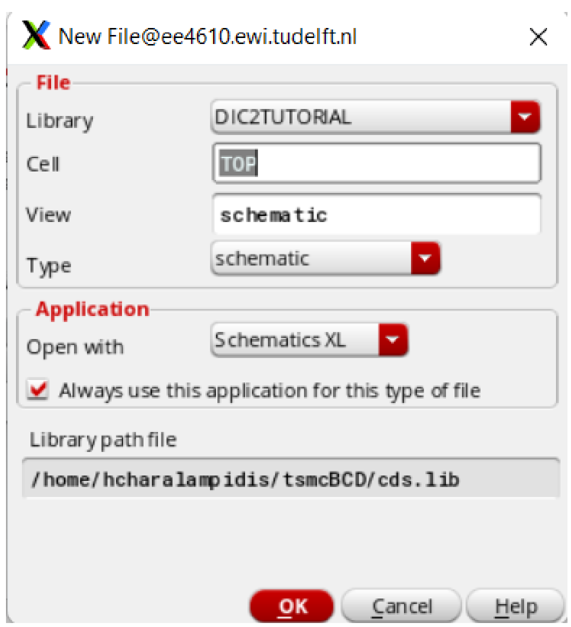
9. Make sure your new library is highlighted. Go to File → New → Cell View.



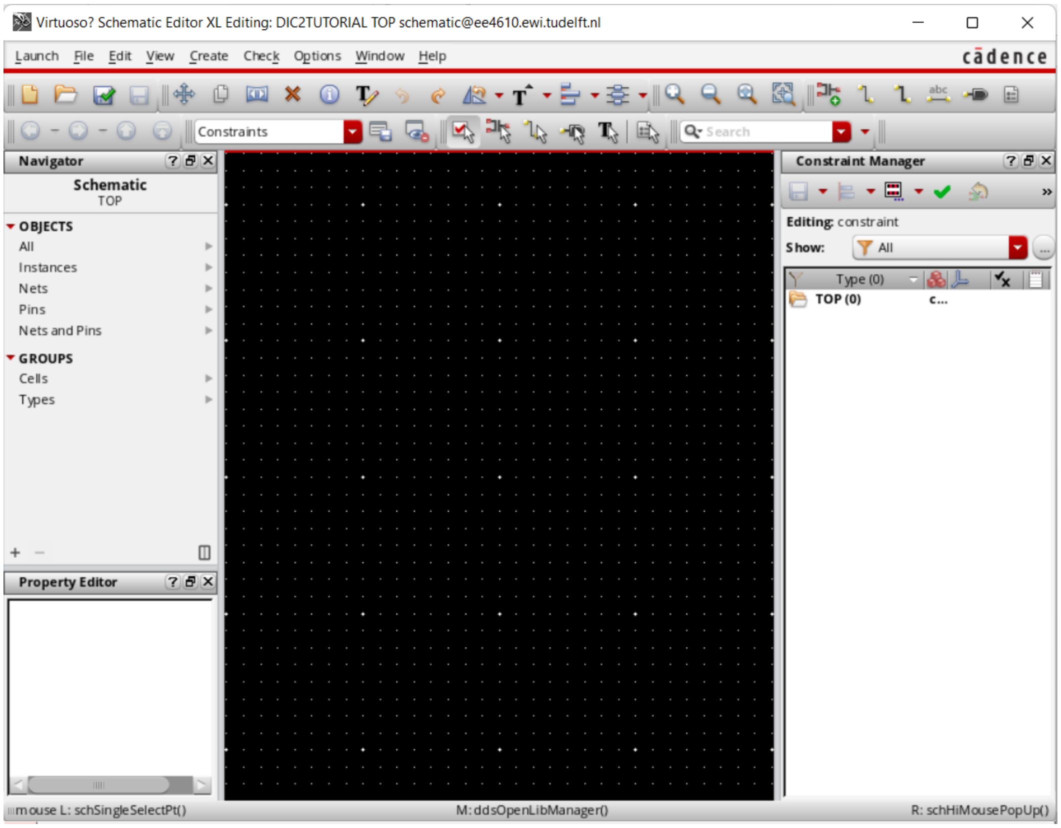


10. On the pop-up window, pick a name for your top-level schematic. As you will be mainly working within this schematic, we suggest a name such as "Main", "TopLevel", "Full", etc.

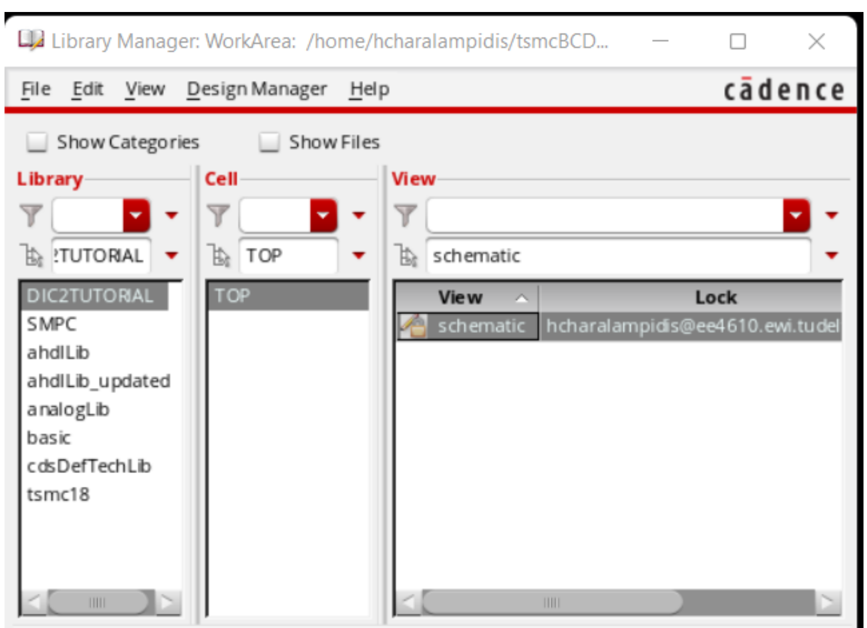
Make sure that your library is selected, and that the "Type" and "View" fields are set to "schematic. For "Open with", pick "Schematics XL", and check the "Always use…" field.



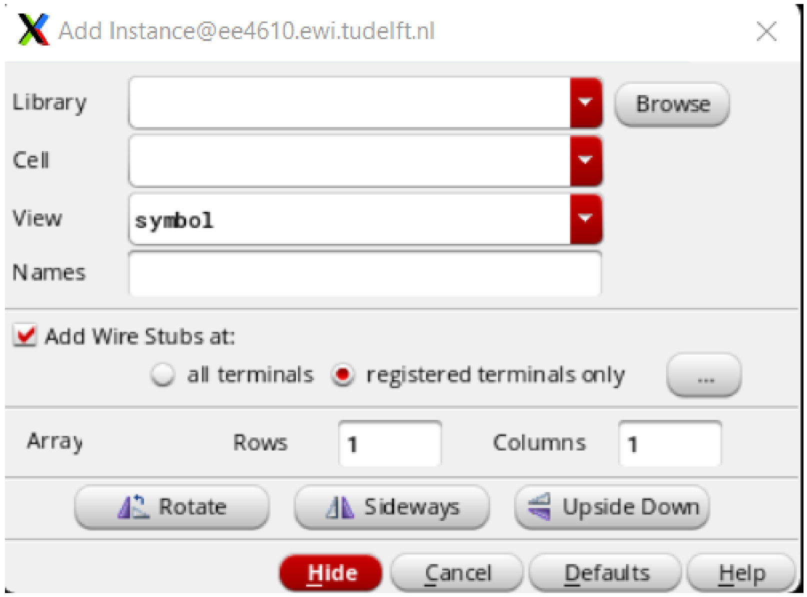
11. The following window will pop up. This is the schematic editor. Feel free to close the side panels for now.



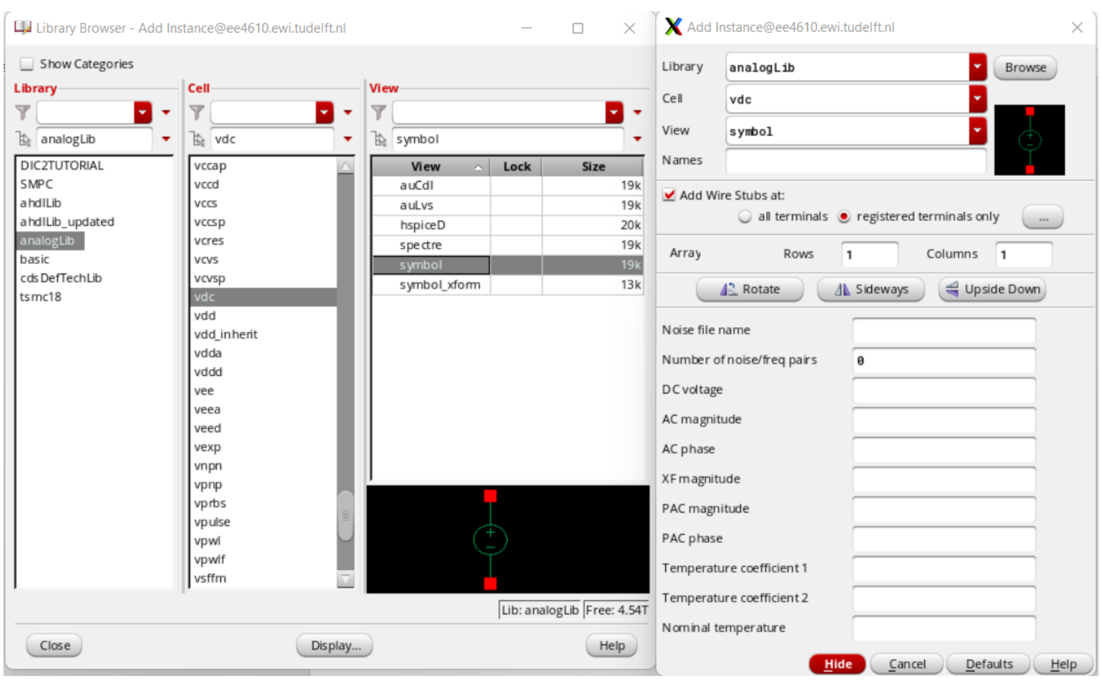
If closed, the schematic editor can be opened through the Library manager. Navigate to your schematic by using the three panels: “Library” → “Cell” → “View” and double-click on “schematic”.



12. To insert a component, press “i” while in the schematic editor window. The following will pop up.



13. Click “Browse” next to the “Library” field. The “Library Browser” window will open:

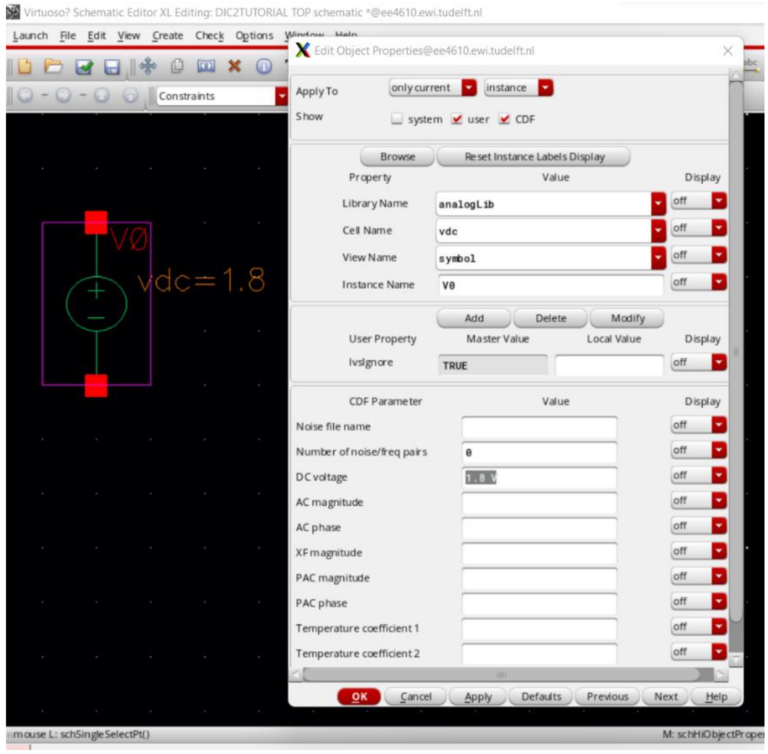


14. Relevant libraries from the left-most panel are “analogLib” which contains mostly common or ideal components, and “tsmc18” which contains technology models such as the transistors used for the project. To add a DC voltage source, navigate towards analogLib/vdc/symbol. You should now be able to click inside your schematic, and an instance will be placed.

15. You can select instance properties such as DC voltage from the “Add Instance” window before placing it, or afterwards by clicking on the instance inside the schematic editor and pressing “q”.

    Set “DC Voltage” as 1.8. and click “OK”.

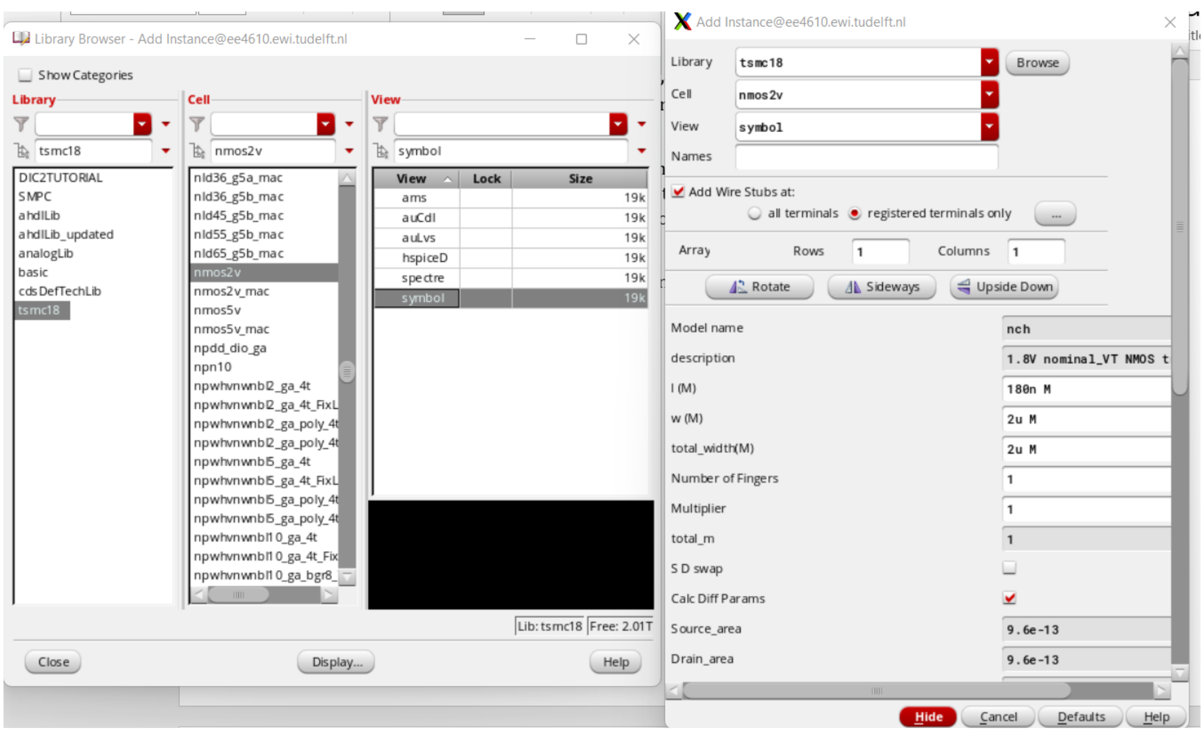
    Note that **not** all property fields need to be filled in for the instance to work as there are default values that will be used if you leave a field blank.



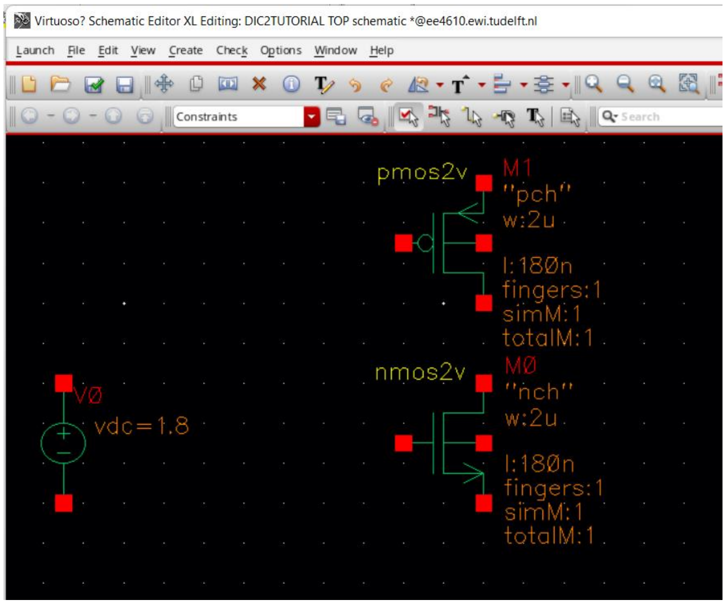
16. To add a transistor, re-open the “Add instance” window by pressing “i", click “Browse”, and navigate to tsmc18/nmos2v/symbol. This is the nominal NMOS transistor used for the project.

    Click inside the schematic editor to place the transistor.

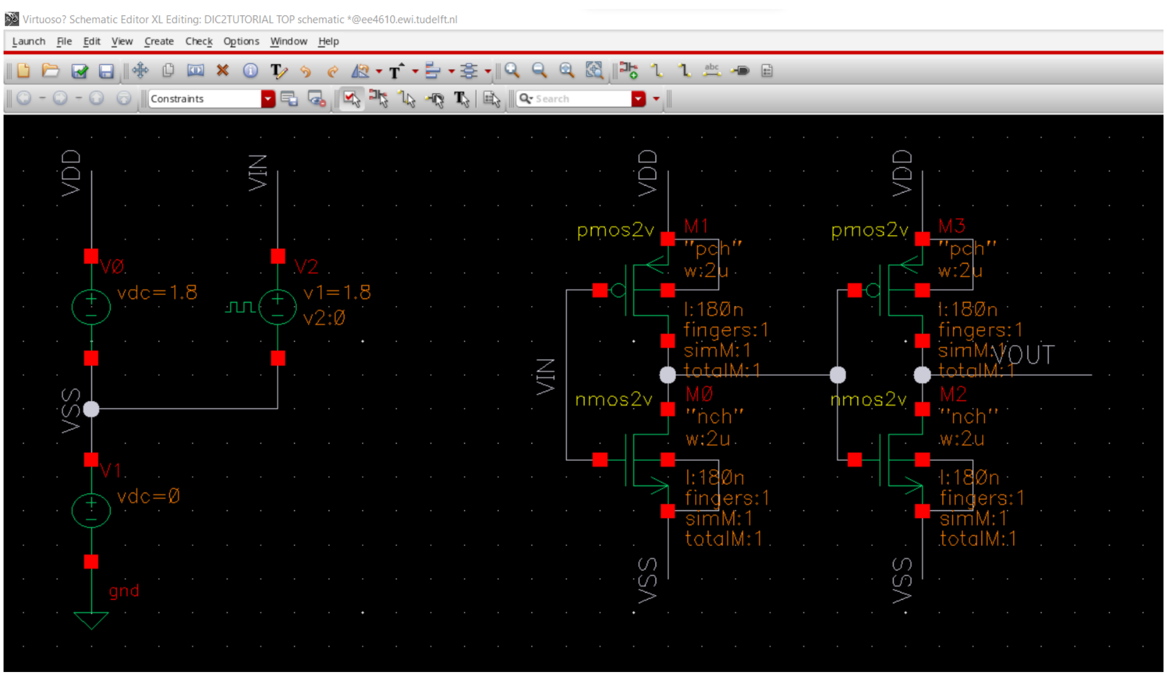
    As with before, you can select the length, width, and number of fingers before placing, and change at any moment by pressing “q” after highlighting the instance in the schematic.



17. To add a PMOS transistor, use tsmc18/**pmos2v**/symbol instead when adding an instance.

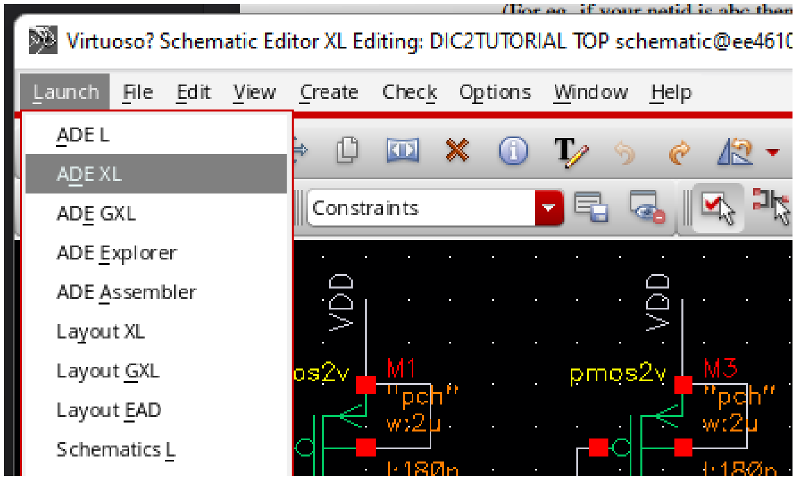


18. You can connect pins to each other by pressing "w" in the schematic editor to select the wire tool. You can name your wires with "L". By giving the same name to different wires, they are considered connected to each other. Check the screenshot below for an example.

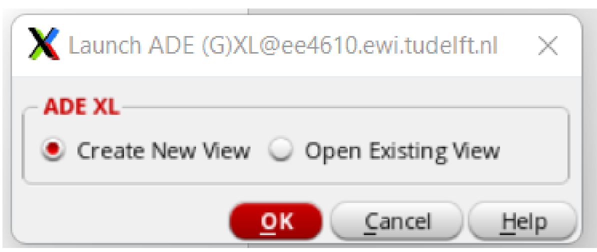


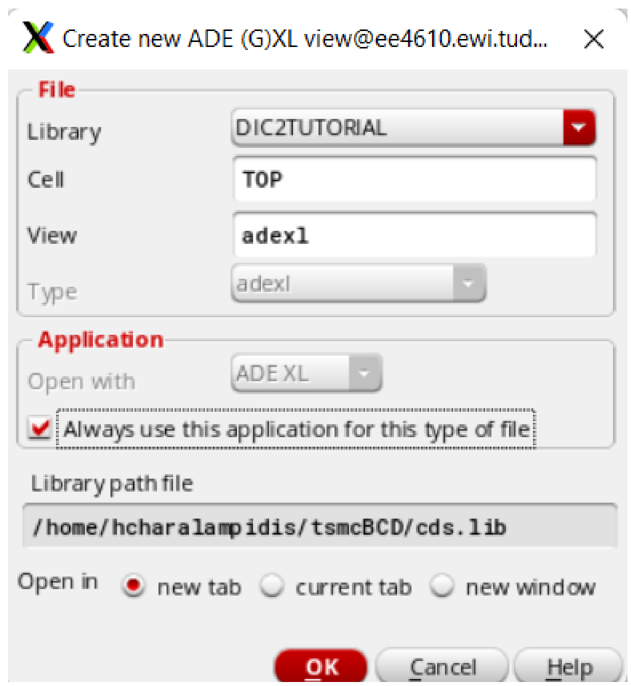
Here, an extra "vdc" instance has been added, along with a ground (instance analogLib/gnd/) as well as a pulse-wave source (instance analogLib/vpulse/) set to 100 MHz. To create the second inverter of this example, select the first one, press "c", and click. You should be able to place a copy of your selection.

19. To perform a simulation, you need to launch a simulation environment. click on Launch → ADE XL. Depending on preference, any of the other ADE options can be used instead.

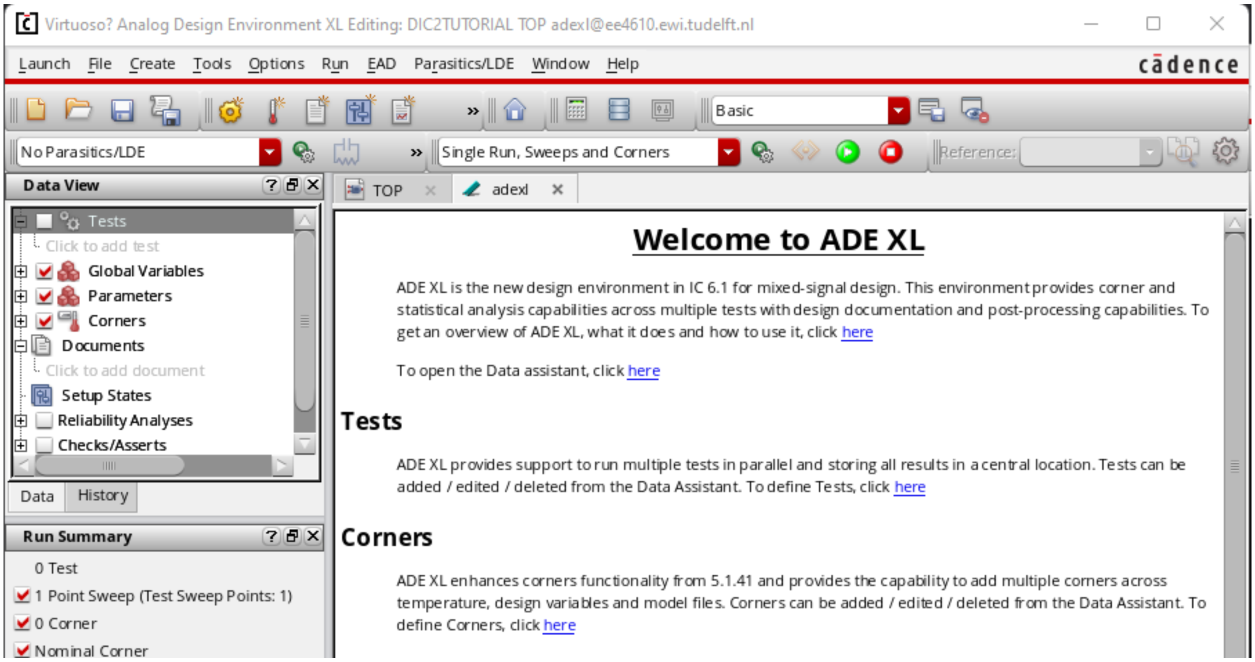


20. The following windows will pop up. Press "OK" on both.

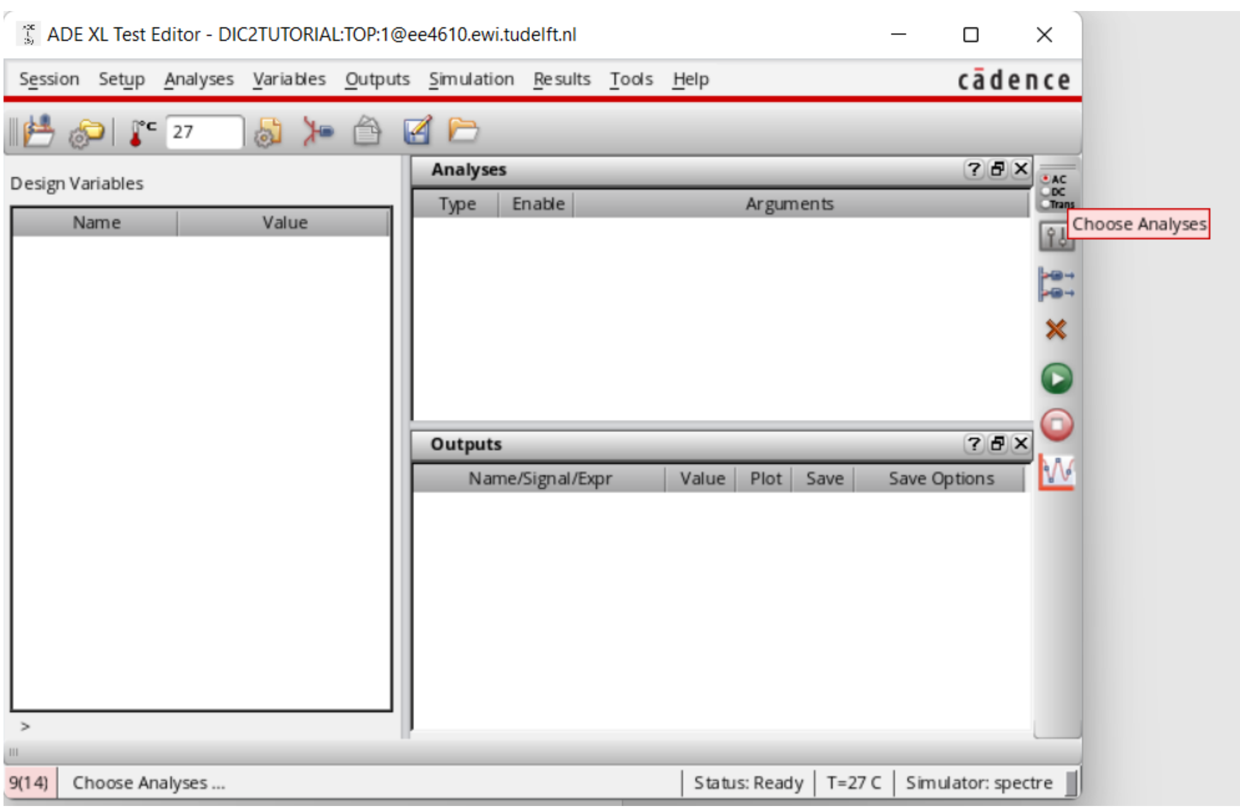




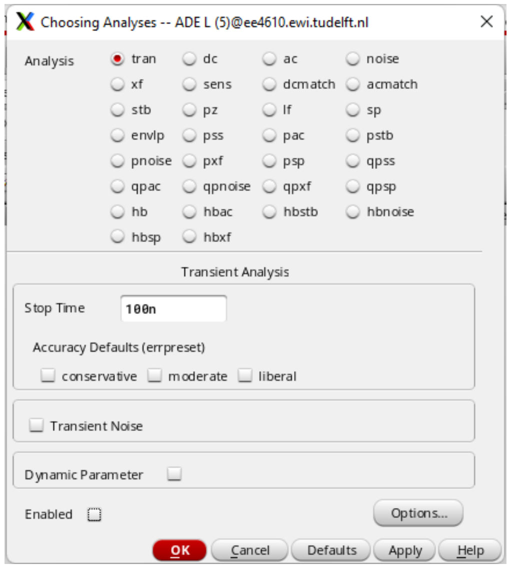
21. The ADE XL window will pop up next. This is where the simulations will be set up and performed. You can access this later from the Library Manager as view "adexl" above "schematic" in the main cell in your library.



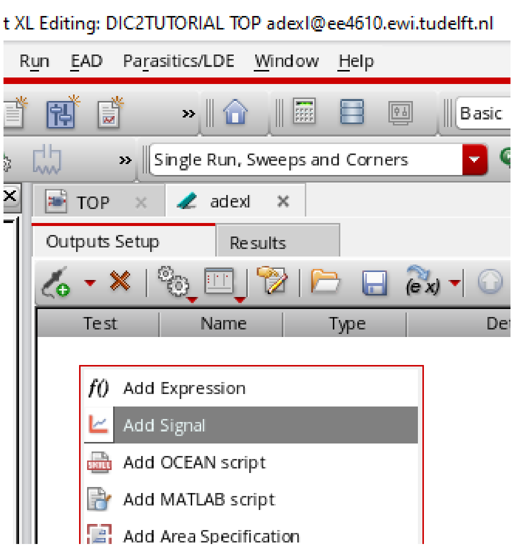
22. To perform a simulation, you need to set up a test first. Click on "Click to add test" on the left-most panel. The "ADE XL Test Editor" will pop up.

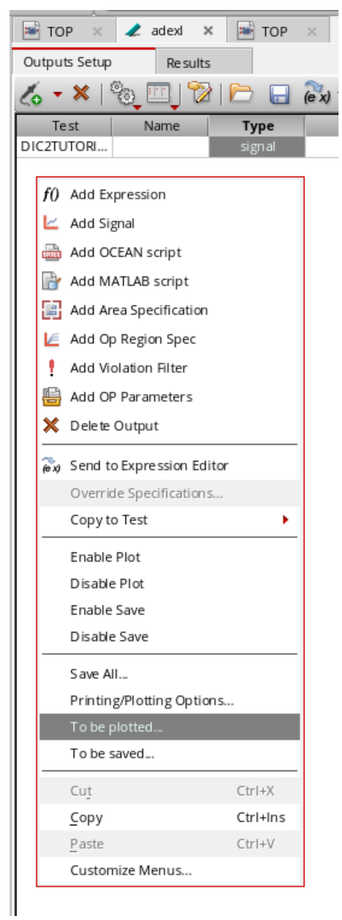


23. Click on the button for “Choose Analyses”. From the pop-up, you will be able to select the analysis type and its parameters. For this example, let’s choose “tran” for a transient analysis, and set “Stop Time” to 100ns. After pressing “OK”, close the Test Editor window.



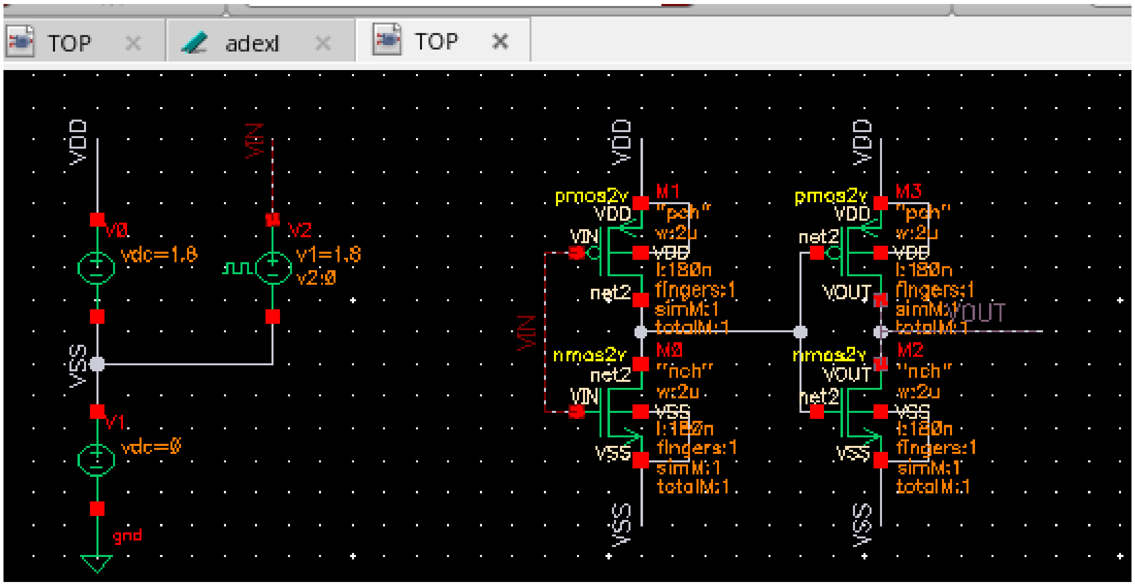
24. The next step is to set up your outputs. These are the signals that you want to plot after our simulation. In the ADE XL window, under tab “Outputs Setup” , right-click and select “Add Signal”.





25. A table row will be generated. Right-click on any cell and select “To be plotted”.

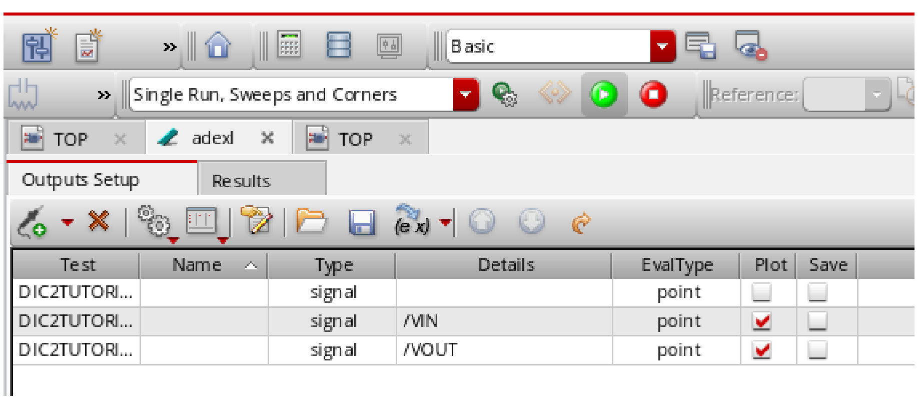
26. Your schematic window will open in a new tab. On there, click on each node that you want to plot. For this example, let’s click on wires “VIN” and “VOUT”. Afterwards, hit “Escape” and return to the “adexl” tab. Your two outputs should be visible as new table rows.



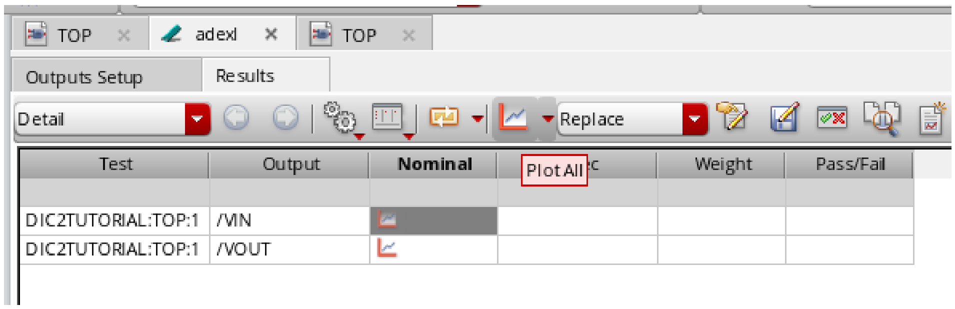
27. Return to the schematic editor and press “Shift + X” to check & save your schematic. Any errors will be described in the main Virtuoso window from step 1.

Note that **not** all warnings need to be fixed. In many cases the warnings will not cause any problems, although it is good practice to check regardless.

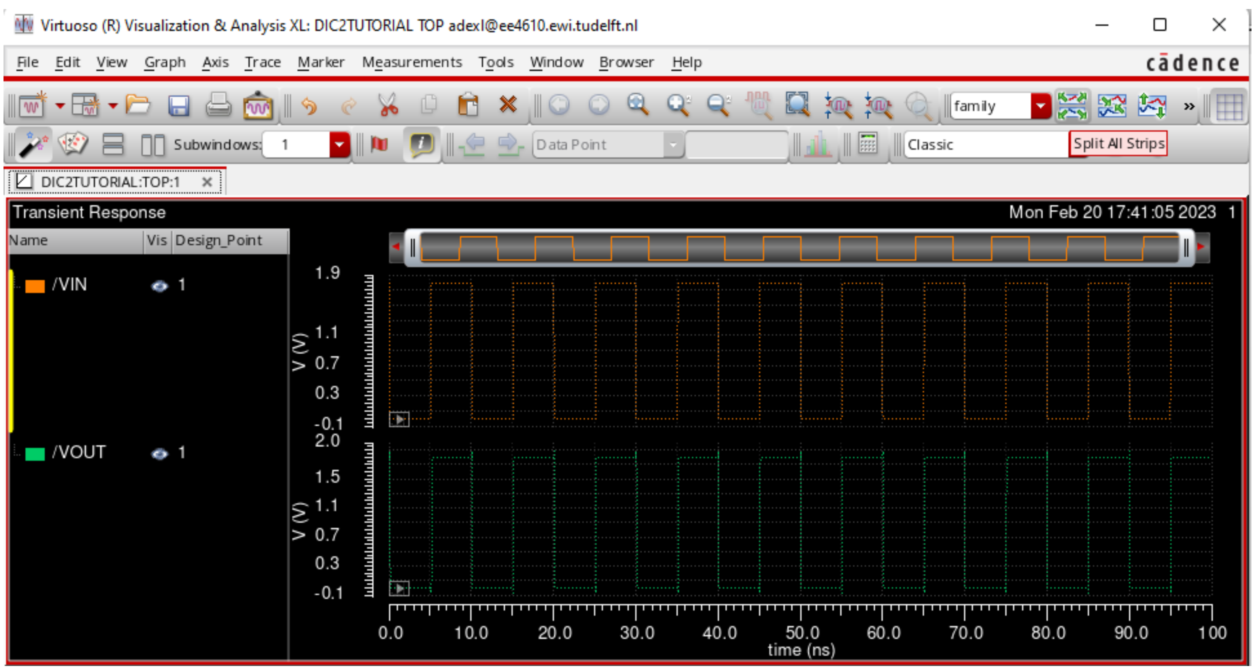
28. Run the simulation by clicking the green “Run Simulation” button in ADE XL.



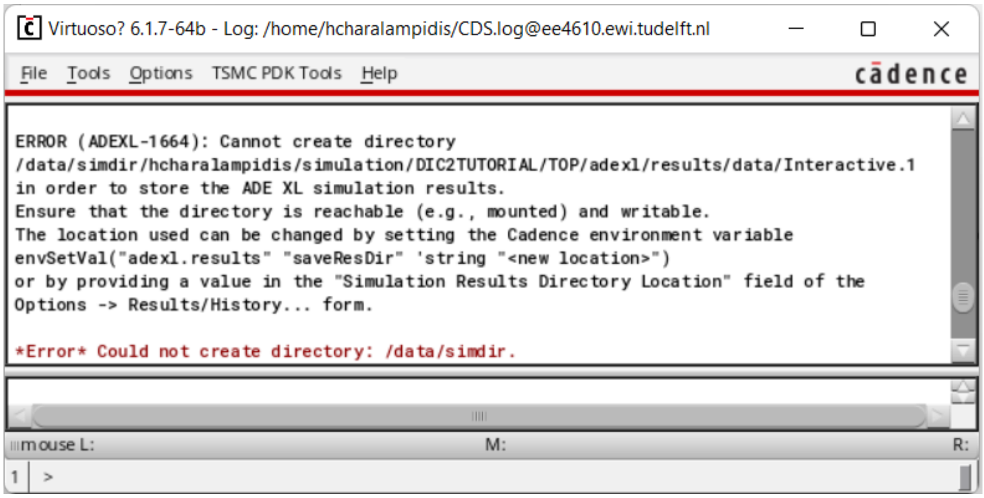
29. After the simulation is over, the "Results" tab will activate. From there, each of the outputs can be plotted separately or by using the "Plot All" button.



30. Creating a plot opens the Visualization window showing a trace for each output. We suggest that you also click the "Split All Strips" button in the case of multiple waveforms.

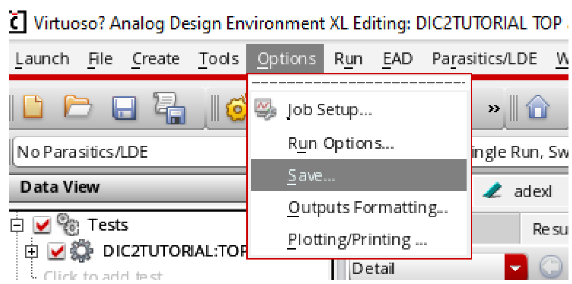


Storage Error: If clicking on the “Run Simulation” button does not seem to work, check the main Virtuoso window for an error such as the one below:



If this happens, change your simulation folder to a local directory with the following steps:

1. ADE XL window → Options → Save.
2. “Save Options” window → Results Location → Browse.
3. “Select Directory” window → Up one level to directory /home/(your NetID).
4. Create a new folder for simulations (e.g. “sims”) → Choose.



5. The “Run Simulation” button should now simulate properly.